SMART ATTENDANCE AND CLASSROOM ANALYTICS

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled **"Smart Attendance and Classroom Analytics"** is the bonafide work of **GANESAN G (230701089), GOKULA SARATHY P S (230701095), HARSHAVARDHANAN V (230701108), HIMESHWAR N (230701116)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

"This project addresses the following Sustainable Development Goal(s): Goal – SDG 4: Quality Education."

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ABSTRACT

Attendance, particularly in the classroom and training management context, is an important but time-consuming activity. The paper introduces the Smart Classroom Attendance and Predictive Twin Analytics System, which implements QR-based and facial recognition-based attendance while also providing attendance management and predictive analytics. The system ensures that attendance is associated with classes and timetables so that the records can be maintained for a particular academic session. QR-based attendance is implemented using time-bound session specific challenge response mechanism protected with HMAC-SHA256 along with institutional authentication and session validation. The facial attendance utilizes a face detection and ArcFace-based embedded face recognition network with cosine-similarity for identity verification. The anti-spoofing tests are also performed for facial recognition attendance. From the recorded attendance data, the system performs attendance analytics including attendance trends, shortages analysis, recovery analysis, and future attendance prediction. Additionally, the system provides what-if attendance analysis and assessment-attendance relationship analysis using Digital Twin and Pearson's correlation analysis, respectively. The functional requirements of the proposed system were tested for the developed attendance management and predictive twin analytics system. The session, QR, facial recognition, security, and analytics modules were found to be operating as designed. The facial recognition module was benchmarked on a CPU and GPU, and the results indicated that the GPU achieved 52.77 ms/frame processing time with 18.95 FPS, compared to the CPU's 1567.63 ms/frame and 0.64 FPS. In other words, the GPU processed 29.70× faster than the CPU. The CPU and GPU achieved similar identity decisions for the benchmark tests. The study concluded that it is possible to design and implement an automatic attendance verification and management and predictive analytics system for a classroom or training management system.

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LIST OF ABBREVIATIONS

Word / Abbreviation	Full Form
API	Application Programming Interface
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
FPS	Frames Per Second
GPS	Global Positioning System
GPU	Graphics Processing Unit
HD	High Definition
HMAC	Hash-based Message Authentication Code
HOG	Histogram of Oriented Gradients
IEEE	Institute of Electrical and Electronics Engineers
ID	Identification

Word / Abbreviation	Full Form
JWT	JSON Web Token
L2	L2 Norm (Euclidean Norm)
MUI	Material UI
OIDC	OpenID Connect
OLS	Ordinary Least Squares
ONNX	Open Neural Network Exchange
OTP	One-Time Password
PAD	Presentation Attack Detection
QR	Quick Response
REST	Representational State Transfer
RFID	Radio Frequency Identification
SAMS	Smart Attendance Monitoring System
SHA-256	Secure Hash Algorithm 256-bit
SMS	Short Message Service
TC	Test Case
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Attendance is one of the key aspects of managing classroom or training sessions at schools, colleges, training institutes or other organised training centres. It provides the record of the participants and helps keep track of attendance of faculty or trainees throughout the course or training session. However, manual marking and maintaining the attendance records can cost a lot of time and effort, especially with multiple sessions and participants.

Automating attendance can ease the process by implementing electronic marking with the help of QR codes or facial recognition technology. An efficient attendance system should also maintain the attendance records with reference to the corresponding class or session and provide the required attendance information for monitoring, along with providing the facility to identify and address attendance shortages. Besides, the attendance records can be analysed along with the academic performance to determine their relationship.

The proposed project, Smart Classroom Attendance and Predictive Twin Analytics System, addresses these aspects with automated attendance and session management as well as attendance and academic performance analytics in one system.

1.2 PROBLEM STATEMENT

Attendance marking is an everyday activity in classrooms and training programmes conducted in schools, colleges, training institutes, and corporate training environments. The traditional attendance methods involve manual efforts which can be tedious when attendance is marked and maintained for multiple sessions. An automated method is required to reduce the time and effort required in managing attendance.

The system must facilitate automated attendance through suitable methods like QR codes and facial recognition. The faculty or trainers must be able to obtain attendance reports for their sessions and the students or participants must be able to monitor their attendance percentage and any attendant shortages.

Attendance data can be utilized for further analysis. By examining attendance in the context of academic performance, information about their observed relationship can be gleaned. In training environments, attendance information can also be used for making decisions concerning assessments and completion of training. Hence, the project addresses the need for an integrated system which includes automated attendance marking, attendance monitoring, reporting and attendance based analytics for classroom and training environments.

1.3 OBJECTIVE

The general objective of the project is to develop a Smart Classroom Attendance Twin Predictive Analytics System which automates attendance marking process, attendance monitoring and analysis in classroom and training.

The specific objectives include:

1. Automatize the process of marking attendance as way of reducing time and effort involved in marking and recording attendance.

2. Implement QR-based attendance process by having time bound and session attendance verification QR.

3. Implementing facial-recognition based attendance for verifying students or participants identity for attendance marking process.

4. Maintaining attendance records according to classes and sessions as a means of having organized attendance data.

5. Monitoring attendance and shortage information to assist students or participants to track their attendance status.

6. Analysis attendance trends and predict future attendance to help in attendance planning.

7. Analyzing the relationship between attendance and assessment results using statistical analysis.

8. Providing attendance reports and analytical information for the faculty, trainers and other relevant users of the data.

1.4 EXISTING SYSTEM

In the traditional settings of classroom and training environments, attendance is marked manually by the faculty or trainers. This requires the presence of each individual student or participant to be recorded in relation to each individual session or module and stored for subsequent use. When there are multiple participants and/or repeated sessions, the effort required for such a process is extremely high and tedious.

Attendance marking manually also provides minimal support for monitoring as well as analysis. Attendance percentages, shortages, session-wise records and reports will require further manual intervention. Thus, the attendance record is primarily just a record of attendance, and does not act as an integrated source of monitoring and analysis.

Automated approaches to attendance marking can reduce this effort, with the use of QR codes, facial recognition, RFID, OTP-based attendance, etc. However, the needs of the proposed system go beyond marking attendance, as it needs to be tied to the relevant classes/sessions, monitored, reported and then used for analysis with regards to the assessment scores. Thus, an attendance system in a classroom or training environment will require more than just automated attendance marking, and will need integrated attendance management and analysis.

1.5 PROPOSED SYSTEM

The proposed Smart Classroom Attendance and Predictive Twin Analytics System aims to automate marking attendance and support attendance recording and analysis in the classroom and training setting. The implemented system makes use of QR-based verification and facial recognition to mark attendance. Records are kept for the relevant class and session for purposes of monitoring and management.

For QR-based attendance, the system makes use of time-bound, session-specific challenges coupled with HMAC-SHA256, institutional authentication and session validation. For facial-recognition based attendance, the system makes use of face detection, and ArcFace-based 512 dimensional facial embeddings for identity verification through cosine similarity. The system also incorporates anti-spoofing checks for the facial attendance process.

The system further analyses the attendance records to determine trend attendance records, shortages, recovery requirements and future attendance projections. The component of the Digital Twin supports what-if analysis based on attendance, while Pearson correlation is used to examine the observed relationship between attendance and assessment results.

The proposed system therefore aims to integrate automated attendance verification, session attendance management, monitoring, reporting and analytics into a single system. It goes on to extend attendance management beyond marking attendance by providing structured attendance information for monitoring and further analysis.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

2.1.1 FACE-RECOGNITION-BASED ATTENDANCE

Face recognition has been considered for automating the classroom attendance process. Bhattacharya et al. proposed Smart Attendance Monitoring System (SAMS) based on face detection, face quality assessment, and face recognition to achieve classroom attendance [1]. Chintalapati and Raghunadh suggested an automated attendance system comprising image acquisition, face detection, preprocessing, feature extraction, classification, and post-processing stages. They also highlighted the spoofing challenge associated with such systems for attendance management [2]. Dev and Patnaik evaluated a face-recognition-based attendance system for variations in illumination, head motion, facial expression, and distance between the camera and the subject to demonstrate image acquisition conditions that can affect recognition outcomes [3].

Several studies have focused on real-time recognition for classroom attendance. Sultan et al. presented a university classroom attendance system based on an HD camera, HOG-based facial feature extraction, and Dlib face-recognition API that demonstrated an accuracy of 97.38% in their experiments [4]. Lateef and Kamil proposed a smart-classroom attendance system based on YOLOv7 and evaluated the performance of conventional machine learning and deep learning-based face recognition algorithms using a database comprising 31 students' images [5]. Raj et al. also suggested a face-recognition smart attendance system that minimizes the manual attendance mark [6]. In another study, multimodal verification was considered to improve the performance of automated attendance systems. Akbar et al. demonstrated an attendance management system

comprising face recognition and RFID-based verification for classroom applications, which suggests using two separate identification methods for achieving better results [7]. However, such a solution would also require additional hardware for RFID-based identification.

2.1.2 QR-CODE-BASED ATTENDANCE

QR Codes have been proposed as a mobile technology for attendance recording. Liew and Tan suggested a QR-code-based student attendance system that incorporated class timings, registered devices, and location data, along with the QR scan [8]. Wiriasto et al. proposed an Android attendance application using Flutter with student and lecturer modules that included location-based QR-scan [9]. Bacuna and Dadiz put forth an idea for eSAM, a QR-based attendance system that could record course, section, subject, date and attendance information providing summary reports for the instructor and administrators [10].

Other researchers have added features of tracking and management in their QR-based attendance systems. For example, Casunuran et al. designed a QR attendance system that could provide GPS tracking with SMS updates for student arrivals and departures [11]. Masalha and Hirzallah introduced a QR-based student attendance management system [12], while Nuhi et al. developed a web QR system for attendance recording and automatic generation of attendance information [13]. Pati et al. suggested a QR-based attendance tracking system with unique QR codes and the recording of student entry and exit information [14]. Ambaye et al. designed and evaluated an automated QR attendance system for classroom management [15].

The researchers describe several QR code-based attendance management systems. Their applications include features like time and date stamp, QR-based entry and exit with tracking and SMS updates. The systems have been reported to reduce record-keeping efforts and increase user acceptance.

2.1.3 DISCUSSION OF THE REVIEWED LITERATURE

The reviewed articles describe the use of facial recognition and QR codes for automating the process of attendance tracking. The face-recognition-based methods emphasize the identification of a student using their biometric data, while QR-based solutions offer a mobile-friendly way of recording the attendance. The literature also reports on the presence of other verification checks, such as spoofing algorithms, registered device and location check, RFID verification, tracking, and attendance reporting mechanisms.

However, the papers do not address the automation of the entire process of student attendance in classes. The current project aims to combine both face recognition, QR attendance, and session management in one educational platform. The platform also introduces the aspects of institutional authentication, presentation-attack spoofing detection, attendance reporting, and analytics. The identified gap in the literature and how the proposed research addresses it is described in 2.3 and 2.4.

2.2 COMPARISON AND DISCUSSION

The reviewed studies explore automated attendance utilizing diverse verification mechanisms and additional functions. For the current project, the literature can be evaluated primarily according to the attendance verification method, additional validation mechanisms, and attendance-management scope. Table 2.1 presents a comparison of the reviewed literature based on these key aspects.

Table 2.1. Comparison of Reviewed Literature

Reference	Primary Method	Additional Mechanisms / Functions	Main Scope
Bhattacharya *et al.* [1]	Face Recognition	Face detection; face-quality assessment	Classroom Attendance
Chintalapati and Raghunadh [2]	Face Recognition	Image acquisition; preprocessing; feature extraction; classification; post-processing	Automated Attendance
Dev and Patnaik [3]	Face Recognition	Evaluation under illumination, head movement, facial expression, and camera-distance variations	Student Attendance
Sultan *et al.* [4]	Face Recognition	HD camera; HOG-based detection; Dlib recognition	Real-Time University Classroom Attendance
Lateef and Kamil [5]	Face Recognition	YOLOv7; machine-learning and deep-learning approaches	Smart Classroom Attendance
Raj *et al.* [6]	Face Recognition	Automated facial identification	Smart Attendance
Akbar *et al.* [7]	Face Recognition + RFID	Multiple identification mechanisms	Classroom Entry and Attendance
Liew and Tan [8]	QR Code	Class time; registered mobile devices; geolocation	Student Attendance

Wiriasto *et al.* [9]	QR Code	Android/Flutter application; geolocation-based scanning	Student and Lecturer Attendance
Bacuna and Dadiz [10]	QR Code	Course; section; subject; date; attendance records; summary reports	University Attendance Management
Casunuran *et al.* [11]	QR Code	GPS-based tracking; SMS notification	Attendance and Location Tracking
Masalha and Hirzallah [12]	QR Code	QR-based attendance identification	Student Attendance
Nuhi *et al.* [13]	QR Code	Web-based attendance recording; automatic attendance information	Attendance Management
Pati *et al.* [14]	QR Code	Personalised QR codes; entry/exit recording	Attendance Tracking
Ambaye *et al.* [15]	QR Code	Classroom management; reporting	Automated Classroom Attendance

The face-recognition studies primarily focus on biometric recognition methods. Bhattacharya et al. [1] utilized face detection and face-quality assessment for classroom attendance. Chintalapati and Raghunadh [2] presented a vision-based processing pipeline which comprises image acquisition, face detection, preprocessing, feature extraction, classification, and post-processing. Dev and Patnaik [3] explored recognition with varying illumination, head movement, facial expressions, and camera distance. Sultan et al. [4] and Lateef and Kamil [5] evaluated HOG/Dlib and YOLOv7-based real-time classroom-attendance recognition systems, respectively. Finally, Raj et al. [6] also applied face recognition for automated attendance.

Akbar et al. [7] proposed an extended concept that was based on face recognition and RFID. Such a method can be considered multifactor since it includes both face detection and RFID tag verification. In this case, RFID is used as an additional security measure with separate hardware to ensure that only authorized persons enter the building. The QR-based methods primarily rely on mobile QR-scanning technology for attendance. Liew and Tan [8] added the current time, registered mobile phone, and geolocation as additional security factors. Wiriasto et al. [9] proposed an Android-based QR attendance system that also utilized geolocation. Bacuna and Dadiz [10] developed a QR system for recording attendance data that comprised course, section, subject, date, attendance records, and report data.

Some QR methods also offer additional functionalities, such as tracking and management. Casunuran et al. [11] proposed a QR attendance system that comprised GPS location tracking and SMS notifications. Masalha and Hirzallah [12] evaluated QR-based student attendance identification, whereas Nuhi et al. [13] presented a web-based QR student attendance management system. Pati et al. [14] utilized personalized QR codes with registration and exit features. Ambaye et al. [15] conducted a study on QR-based attendance for classroom management. They found that QR attendance improved record keeping, reduced paperwork, and was generally well received by users.

Based on the review it can be concluded that the existing QR- and face-based attendance systems offer different functionalities and operate on different principles. Whereas face-recognition methods primarily focus on addressing issues of identity recognition, QR methods primarily offer attendance recording with some additional features. Some methods offer additional functionality, such as RFID verification, device and location recognition, SMS, or attendance reports. However, none of the reviewed QR or QR-based attendance methods offer all of these features. The proposed system is designed to offer a session-based approach wherein one QR code is used per session which is cryptographically secured with HMAC-SHA256, institutional confirmation, and session validation. The face-recognition technology utilizes face detection, facial embeddings,

identity matching, and presentation-attack detection. Both methods provide session-based attendance recording, whereas the QR method also offers additional session tracking, SMS alerts, and attendance reports. Thus, none of the existing QR-based attendance methods offer all the features provided by the proposed system.

The analysis of the face-recognition methods also shows that they primarily address the identity-recognition component of the proposed system. However, none of them provide a session-based approach with cryptographically secured QR codes or institutional authentication. Moreover, most face-recognition methods do not offer presentation-attack detection. Therefore, the existing face-recognition methods are insufficient in terms of providing all the features of the proposed system. The next section will discuss the research gap in more detail.

2.3 RESEARCH GAP

The reviewed literature establishes the use of face recognition and QR codes for automated attendance, with individual studies incorporating additional mechanisms such as RFID verification, registered-device validation, geolocation, tracking, notifications, and attendance reporting [1]–[15].

However, the systems do not demonstrate the same attendance verification mechanisms, security controls, and classroom and academic-management functions. In particular, the works do not show the integration of face recognition and presentation-attack detection, time-bounded cryptographically protected QR authentication, institutional identity validation, and classroom and academic management within a system. The gap is therefore not the individual technologies. Face recognition and QR-based attendance have been investigated, and other studies have incorporated additional identification or contextual mechanisms.

The systems are missing the integration of multiple attendance mechanisms and their associated security checks in a common session-based attendance-management

framework. The proposed system addresses this gap by integrating QR-based and facial-recognition-based attendance with classroom session management and common attendance-recording infrastructure.

QR attendance incorporates time-bound challenges, HMAC-SHA256 protection, institutional authentication, and session validation. Facial-recognition attendance incorporates face detection, facial embedding generation, identity matching, and presentation-attack detection.

The project therefore does not claim novelty for the individual face-recognition, QR, or cryptographic techniques. Its contribution is the practical integration of these attendance mechanisms, their associated security checks, and classroom and academic-management functions within a unified platform; and attendance reporting and analytics operate on the same records.

2.4 PROPOSED SYSTEM RESPONSE

The proposed system addresses the identified gap by incorporating a variety of attendance-verification methods in a common session-based attendance-management system. The proposed system provides QR-based attendance and facial recognition-based attendance as an alternative method for recording attendance, linked to the relevant classroom session.

For QR-based attendance, the system implements a time-bounded, rotating challenge with HMAC-SHA256 protection. The attendance request is additionally verified by institutional Google authentication, session validation, and section validation prior to attendance record creation. This allows the QR attendance process to authenticate identity, session, and QR challenge rather than only the QR code.

For facial-recognition-based attendance, the system implements a pipeline of face detection, frame filtering, facial embedding generation, identity matching and presentation-attack detection. The MediaPipe is used for initial face detection, whereas InsightFace buffalo_l is used to generate 512D L2-normalized facial embeddings, with cosine

similarity for identity matching. MiniFASNetV2 and MiniFASNetV1SE are used for presentation-attack detection, along with additional image-based checks.

Both attendance methods are linked to classroom session and timetable management and share a common attendance-recording framework. This enables the system to maintain attendance records and deliver attendance reports, shortage information and attendance analytics. The attended records are further used for trend analysis, future attendance projection, Digital Twin-based what-if analysis and examination of reported observed relationship with assessment results.

Thus, the proposed system responds to the identified research gap through the practical incorporation of attendance verification, security validation, session management, reporting and analytics in a unified classroom and academic-management platform. The contribution is the incorporation of the components rather than developing a new face-recognition algorithm, QR technology. cryptographic protocol.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed system comprises a client-server architecture that accommodates a frontend, backend services, attendance verification processes, and a PostgreSQL database. The frontend is developed using React and Vite, whereas the backend is implemented using Django and Django REST framework. PostgreSQL is used for persistent storage of the application and academic data. The overall architecture of the proposed system, including the client interfaces, backend services, attendance verification pipelines, data layer, and analytics modules, is illustrated in Fig. 3.1.

The application's frontend accommodates faculty/teachers and students. Whereas the faculty manages classroom sessions, attendance, reports, and academic information, the students access their attendance records and their related academic information. The Django REST backend accommodates the application's central services. It manages the application's authentication, classroom sessions, attendance, academic records, and analytics. Attendance is implemented using two verified attendance methods: QR-based attendance and facial-recognition-based attendance. Both of which connect to the same session-based attendance infrastructure hence facilitating consistent attendance records.

The QR and facial-recognition processes comprise the main attendance-verification paths. QR attendance involves time-bound, session-specific verification with institutional authentication. Whereas, facial attendance involves face detection, facial embedding-based identity verification, and presentation-attack detection. Processing of both QR and facial attendance methods are detailed in Chapter 4. The stored attendance data is connected with class sessions, timetable information, academic records, reporting, and analytics. This architecture facilitates attendance records to support monitoring, shortage identification,

trend analysis, future attendance projection, Digital Twin-based what-if analysis, and the observed relationship between attendance and assessment results.

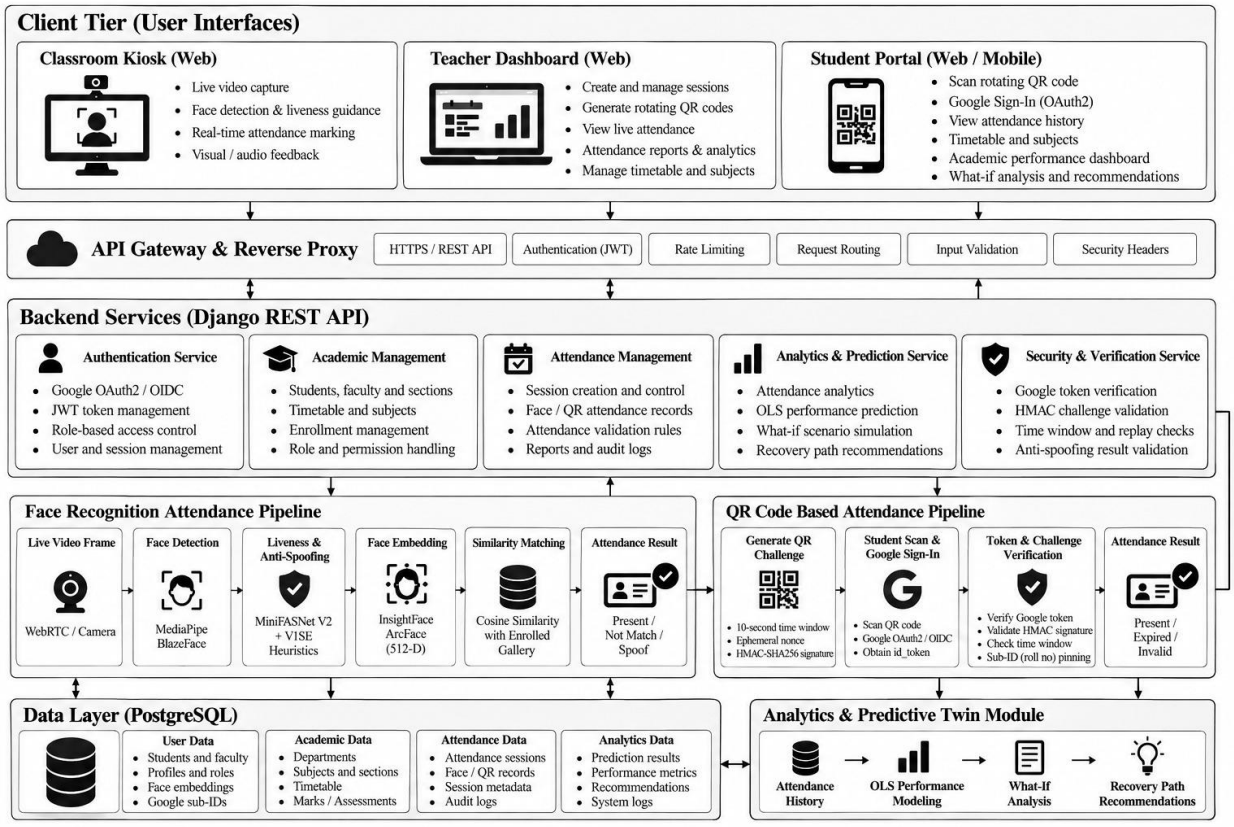


Fig. 3.1. System Architecture

3.2 SYSTEM REQUIREMENTS

The system requirements are organised into software, hardware, data, and deployment requirements. They include only the resources required by the implemented system and avoid detailed implementation parameters that are discussed in later chapters.

3.2.1 SOFTWARE REQUIREMENTS

The software components required for the system are:

- React 19 with Vite 8.1.1 — frontend development.
- Material UI (MUI) 9.2.0 — user-interface components.
- Django 5.0.1 — backend application framework.
- Django REST Framework 3.14.0 — REST API services.
- PostgreSQL — relational database management.
- SimpleJWT — application authentication.
- Google OAuth2/OIDC — institutional authentication for QR-based attendance.
- MediaPipe BlazeFace — client-side face detection.
- InsightFace buffalo_l — facial embedding generation.
- ONNX Runtime 1.27.0 — machine-learning model execution.
- OpenCV — image processing.
- NumPy and SciPy — numerical and scientific processing.

Together, these requirements support the implementation of user interfaces, backend services, attendance verification, data management, and analytical functions of the system.

3.2.2 HARDWARE REQUIREMENTS

The attendance methods implemented require appropriate input and computing devices to operate. Facial-recognition attendance is conducted using a computer with a camera enabled, and QR-based attendance requires a student's mobile device with a camera to scan.

The pipeline for processing faces can leverage a CUDA-compatible GPU, accelerated through the CUDA execution provider, resulting in increased processing speed. CPU is used as the fallback processing environment if a GPU is unavailable.

The project documentation does not specify minimum processor, memory, or GPU requirements. As such, particular hardware specifications are not identified as mandatory requirements.

3.2.3 DATA REQUIREMENTS

The system requires following categories of data

- User and student data that could be used to authenticate the user and know the student.
- Class and section data to manage students and attendance.
- Timetable and session data to match attendance with corresponding class session.
- Attendance data which includes status, timestamp, verification method, verification metadata.
- Facial enrollment data for facial identity verification.
- QR session data required for QR based session validation.
- Assessment and marks data for academic performance analysis.
- All this data is used for attendance recording, monitoring, reporting and subsequent analytics.

3.2.4 DEPLOYMENT REQUIREMENTS

The system requires the React frontend app, Django backend, and PostgreSQL database to be up and running as the key elements of the application. The network connectivity should be available for the frontend to communicate with the backend and perform authentication-dependent operations.

Facial attendance also requires browser access to a camera, while QR attendance needs a camera-equipped mobile device to be able to scan. Additionally, if there is a CUDA environment available, a GPU can be used to accelerate facial processing; otherwise, the facial-processing pipeline is executed using the CPU provider.

3.3 SYSTEM DESIGN

The system design below is the system's process of the activities performed by users, frontend, backend, attendance-verification module, and PostgreSQL. It explains how the user initiates the attendance process, records the attendance information, verifies it, and later monitors and analyses the data.

3.3.1 ACTIVITY DIAGRAM

The operation process starts with the user's authentication and the availability of functions relative to the corresponding user role. Faculty members are entitled to manage classes and sessions, perform attendance procedures, and store attendance and academic information. In turn, students are allowed to access their attendance and related academic information.

To mark attendance, the user is required to choose the necessary active class or session and apply one of the implemented attendance marking methods. The system processes the selected information and performs the needed verification. In case of QR-based attendance marking, the system verifies the attendance by scanning the QR code and cross-referencing the information with the user's credentials and session data. Meanwhile, facial recognition

procedures require facial information scans for identification and presentation attack detection. After the successful scan, the system stores the collected attendance data relative to the corresponding student and session.

The stored information can subsequently be applied by the user to monitor, report, or analyze the retrieved data. The overall sequence of activities involved in the attendance process, from session initiation to attendance storage, analytics, and reporting, is illustrated in Fig. 3.2. Generally, the described operation sequence demonstrates a continuous flow of operations starting from the user's authentication and attendance marking and concluding with the use of attendance data.

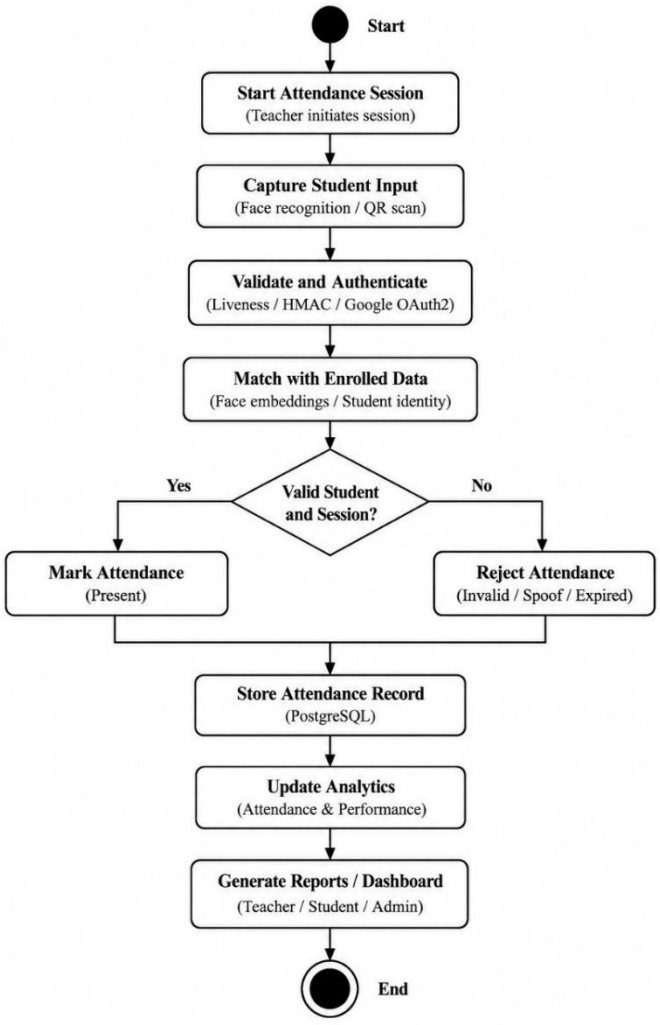


Fig. 3.2. Activity Diagram

3.3.2 DATA FLOW DIAGRAM

The data flow diagram illustrates how information moves through the proposed system. Users interact with the React frontend, which communicates with the Django REST backend, while PostgreSQL manages the storage and retrieval of application, attendance, class, and session data. The overall flow of information between the user interfaces, verification processes, backend services, database, and analytics modules is illustrated in Fig. 3.3.

During attendance marking, information is processed through the appropriate verification method. QR-based attendance uses authenticated identity and session-specific QR information, whereas facial-recognition attendance uses captured facial information for verification. The verified attendance data is then stored against the corresponding class session and utilized for attendance monitoring, reporting, trend analysis, future attendance projection, Digital Twin-based what-if analysis, and assessment-related analysis.

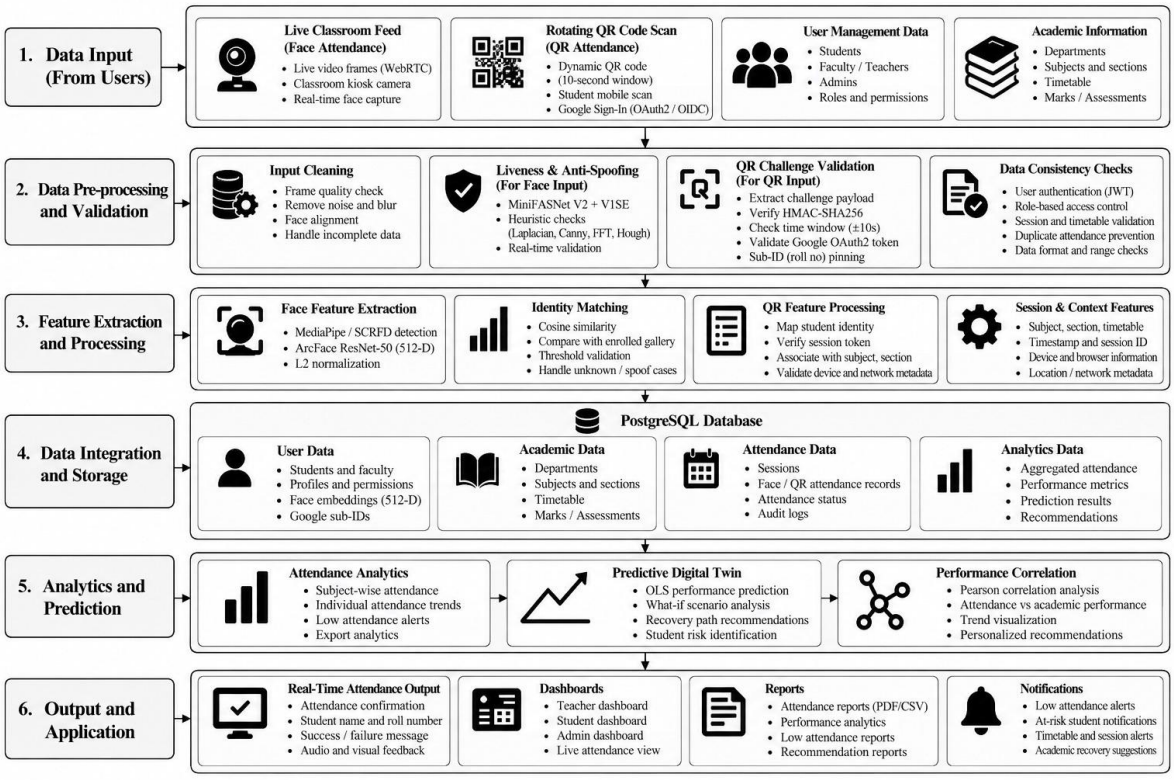


Fig. 3.3. Data Flow Diagram

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The system implements an automated attendance mechanism where the attendance is captured against the specific class and the session. There are two attendance methodologies namely QR-based and facial recognition-based attendance. Both attendances follow a similar process, but the two methods differ in the way the attendance is captured.

4.1.1 AUTOMATED ATTENDANCE

The attendance process is initiated by an active class session that allows attendance. The student then selects the QR-based or the facial recognition-based attendance. The selected process then facilitates the request and the backend validates the request and stores the attendance against the session for the student. The overall process can be explained below: Active Session → Select Attendance Method → Verification → Backend Validation → Attendance Recording → Attendance Storage

The QR-based attendance is achieved through the user authentication and the session-specific, temporally valid, QR challenge. On the other hand, the facial attendance requires the system to capture the user face and perform the required presentation-attack detection before the attendance is marked.

Once the attendance has been captured it forms a part of the student's session-wise attendance. The attendance data is then used for the purpose of attendance tracking,

reporting, trend analysis, projection, and other analytics activities discussed in later sections.

4.1.2 FACE RECOGNITION

The facial-recognition approach is used to verify a student's identity based on a captured picture using a camera before recording the attendance. This includes the stages such as face detection, feature extraction, identity matching, and presentation-attack detection as part of the attendance record process.
The process takes the following form:
Camera Capture → Client-side Face Detection → Frame Submit → Server-side Face Processing → Facial Embedding → Similarity Matching → Presentation-Attack Detection → Attendance Record

The process starts at the frontend using MediaPipe BlazeFace to detect the face from webcam stream. With the client's configuration of detection with face position checks in place, an appropriate image frame is submitted to the backend to process.The frames are submitted in intervals and not continuously to reduce data transmission rate thereby saving the resources during the attendance process. The backend uses the InsightFace buffalo_l model to process the faces.

After detecting the face and getting the necessary facial keypoints, a facial embedding is generated using the ArcFace recognition model. The resulting representation from the model is a 512 dimensional L2-normalized vector obtained from the model. The vector is then matched to the ones captured and registered for the enrolled students using cosine similarity.

The process is looking for a match with a similarity higher than the configured threshold.

During enrollment, multiple photos can be captured to generate the embedded vector for the student, whereas the cached embedded data speeds up the process of matching vectors during attendance processing. The matching of enrolled identities is done alongside with the presentation-attack detection process. The PAD process uses MiniFASNetV2 and MiniFASNetV1SE in tandem with image-based checks to verify the liveness and the security of the subject's facial attendance.

With that, the verification process takes into account the identity and the liveness test result for the processing of the facial attendance record. Once the verification processes meet the required criteria, then the backend records the attendance for the student against with the current session. The attendance record is then available for processing and use in common attendance management and analytics processes.

4.1.3 SECURE QR ATTENDANCE

The QR-based attendance approach stores attendance by combining authenticated user identity and a time-specific QR challenge. The approach is designed in such a way that a QR value generated for a session and a time period cannot be used as a generic attendance value for another session or another time period.

The process is as follows:
Institutional Authentication >> Active Session >> Time-bound QR Challenge >> QR Validation >> Session and Section Validation >> Duplicate Check >> Attendance Recording

The user is first authenticated against the institution’s Google OAuth2/OpenID connect flow. The system validates the necessary identity information including the authenticated account, verified email status, allowed domain and institutional account association.

For an active session, the system generates a QR challenge using the session specific qr_secret. The challenge is generated using HMAC-SHA256 in a 10-second time window. The QR value is therefore associated with both the particular session and the current time window.
On scanning the QR code, the submitted challenge is validated by the server. The system accepts the current time window as well as the immediately preceding window allowing the change between consecutive QR values. The authentication value is then compared using a constant-time comparison.

After QR validation, the system checks the session and section information and verifies that attendance has not been recorded for the student. On satisfying the conditions, the attendance record is then created for corresponding student and session.

In this way institutional authentication establishes the user's authenticated identity, while the time-bound HMAC-SHA256 QR challenge provides session and time validation for the attendance request. Note that HMAC-SHA256 is used for generating message authentication and integrity of the challenge, it does not encrypt the QR data.

4.1.4 ATTENDANCE ANALYTICS

The attendance analytics method takes as input the attendance records processed by the previous methods and provides information on the current attendance status, attendance trends, attendance projection, and recovery analysis. The attendance analytics method also contains the Digital Twin process that allows exploring what-if scenarios.

The process can be generally described as follows:
Attendance Records → Attendance Calculation → Trend Analysis → Future Projection → Recovery Analysis → Digital Twin What-if

First, using the attendance records and session data, the current attendance status is calculated to indicate the student's attendance status and percentage. This data can be used to identify any attendance deficits and track attendance progress towards the desired level. Next, a trend analysis is performed consisting of applying Ordinary Least Square Regression to attendance observations. The slope of the regression line is used to estimate the attendance trend, while the forecasted values from the regression are used to predict future attendance. Since the projection is made using historical values, it should rather be considered an analytical opinion on the likely future attendance.

If the current attendance or its forecast is below the desired level, an analysis of the required recovery is performed to understand how much additional attendance is needed to reach the desired level given the number of sessions left.

Finally, the what-if scenarios are explored using the Digital Twin. By changing various input variables, one can observe how different future attendance scenarios would impact the overall attendance and, thus, the student's attendance status. Note that what-if scenarios are analytical tools that do not change the actual attendance records.

Therefore, the attendance analytics process allows analyzing the current attendance, track attendance trends, project future attendance, analyze necessary recovery, and explore what-if scenarios.

4.1.5 ACADEMIC PERFORMANCE ANALYSIS

The academic performance analysis investigates the relationship between the recorded attendance and assessment. From the attendance and assessment data maintained by the system for analysis, it provides an analytical view of the student academic performance in terms of attendance. The process undergoes:

Attendance Data + Assessment Data → Pearson Correlation → Relationship Analysis

With the use of the existing attendance records as well as the corresponding assessment results for analysis in the system, Pearson correlation is applied to investigate the statistical relationship between the two. The resulting value of Pearson coefficient represents the direction and strength of the relationship observed between the two variables in the data. The analysis is utilized to discover patterns of attendance and assessment performance. The result of correlation represents an observed association in the data and is not used to establish any causation between student attendance and academic performance. The result of the analysis adds up to the academic analytics feature and provides additional information in attending to monitoring of attendance and assessment performance.

4.2 MODULES

The proposed system is structured into modules for user management, attendance management, session and timetable, attendance analytics, academic analytics and reporting. These modules make use of the common application and attendance data that is maintained by the backend and PostgreSQL database.

4.2.1 USER MANAGEMENT

The User Management module handles the authentication of the users and enables the role-based access of the application's functionality. The software offers faculty and student accounts that grant access to the corresponding classroom management tools, attendance tools, and other features.

The application utilizes the SimpleJWT library to handle the application's authentication needs. The institutional Google OAuth2/OpenID Connect is used to authenticate institutional users in QR attendance processing stages whereas the custom JWT

authentication is used to access QR scanners or kiosks. Upon authentication, the user gains access to the set of functions available to their role. The faculty, for example, can access classroom management features while students access attendance features. The User Management module thus enables the access control functionality to the other modules.

4.2.2 ATTENDANCE MANAGEMENT

The Attendance Management module handles the records of the students' attendance for the session they are enrolled in. It serves as the shared functional layer through which the QR attendance records or the facial recognition attendance records are processed and stored.

The system processes the attendance records by associating them with the corresponding student and session. The attendance record includes the additional information about the session's status, the time of attendance record, the type of scanner used, and other relevant information for the verification process. The module processes the result of the selected verification step and stores the record if the required verification procedure is fulfilled. The attendance records are stored session-wise and are used to monitor the students' attendance while also highlighting any attendance shortages. The records are also used by the reporting module for the generation of reports regarding attendance trends, projections, recovery analysis, and assessment of the correlation between attendance, and other factors.

4.2.3 SESSION AND TIMETABLE MANAGEMENT

The Session and Timetable Management module handles the timetable configuration and the class sessions for the attendance. It provides the session context required to associate the attendance with the correct class and scheduled period.

The timetable configuration holds the class timing, period duration and break information.

Based on this configuration, the system generates sessions for the applicable classes. The session generation process also takes into account the configured working days and holidays.

In the implemented system, Sundays are excluded from session generation, while Saturdays are treated as working days. Configured holidays are also excluded. The session creation process is an idempotent process, such that an existing session will not be created again when the generation operation is repeated.

The generated session records form the basis for attendance marking and session-wise attendance management. They also provide the session information needed by the QR attendance process for its session-specific verification.

4.2.4 ATTENDANCE ANALYTICS

The Attendance Analytics module makes use of the recorded attendance and session data to support attendance monitoring and planning. It provides the functional interface for analysing attendance after the attendance records have been created. The module provides the following attendance-related information:

- Attendance percentage and current attendance status
- Attendance shortage
- Attendance trend
- Future attendance projection
- Recovery requirements
- Digital Twin-based what-if analysis

Attendance percentage and shortage information are derived from the student's recorded attendance in relation to the applicable sessions. The module also makes use of attendance history to provide trend and future projection information. Recovery analysis provides the attendance requirement for reaching the configured target when a shortage exists. The Digital Twin component provides scenario-based attendance analysis. Different future

attendance conditions can be examined to understand their possible effect on attendance status without changing the actual attendance records. This allows the module to present the current attendance position and possible future attendance conditions within the same analytical view.

4.2.5 ACADEMIC ANALYTICS

The Academic Analytics module makes use of the attendance and assessment data to observe the observed relationship. The Academic Analytics module goes a step further by combining the attendance data with the assessment data maintained by the system.
The Pearson correlation is used to depict the relationship between the two sets of data. The calculated value helps to establish the direction and magnitude of the relationship between attendance and the assessment results.
The information then aids academic monitoring through taking attendance and performance data into consideration. The analysis highlights the associations observed from the data but does not indicate any causal relationships between the two variables.

4.2.6 REPORTING

Reporting module provides the attendance and analytical reports generated from the records maintained by the system. It provides access to attendance information to monitor the classes and student attendance.

The module provides the session-wise attendance records to help monitor teaching sessions. The faculty can view and access the attendance records of their respective classes and sessions while students get access to their attendance records and related information. The reports are generated using the same data used to support the analytical functionalities of the system. This ensures the use of information to generate the analytical insights and

reports consistently. It also ensures that attendance records, analytics, and reporting capabilities are maintained without having to maintain separate datasets.

CHAPTER 5

IMPLEMENTATION, RESULTS AND DISCUSSION

5.1 IMPLEMENTATION

5.1.1 SYSTEM IMPLEMENTATION

The system was implemented as an integrated web application with a React and Vite frontend, Django and Django REST Framework backend, and PostgreSQL database. The frontend provides the user interface, while the backend manages the application services and communicates with the database for storing and retrieving system information.

The frontend uses React 19 with Vite 8.1.1 and Material UI 9.2.0. The backend uses Django 5.0.1 and Django REST Framework 3.14.0 to provide the application and REST services. PostgreSQL is used as the relational database for the system's persistent data.

Authentication is implemented according to the requirements of the system. SimpleJWT is used for application authentication, Google OAuth2/OpenID Connect is used for institutional authentication in the QR attendance workflow, and custom JWT authentication is used for classroom or kiosk access.

The implemented functions are connected through the backend and shared database. Class and session information provides the context for attendance, while the QR and facial-recognition processes supply the verification results used to create attendance records. These records are subsequently available to the attendance monitoring, reporting, and analytics functions.

This implementation provides the common software foundation on which the attendance, session management, analytics, and reporting functions operate.

5.1.2 ATTENDANCE IMPLEMENTATION

The Attendance Management function was implemented around the class session so that attendance is recorded for the appropriate student and session. The system supports QR-based and facial-recognition-based attendance, with both methods using the same attendance-recording process after successful verification.

In the QR workflow, the authenticated user submits the QR challenge for the active session. The backend validates the challenge and then checks the relevant session and section information and whether attendance has already been recorded. When the required checks are satisfied, the attendance record is created.

In the facial-recognition workflow, the captured facial information is processed for identity verification and presentation-attack detection. When the required verification conditions are satisfied, attendance is recorded for the identified student and active session.

The resulting attendance record maintains the information required for attendance management, including the student, session, status, timestamp, method, and relevant verification information. These stored records are then used by the monitoring, reporting, and analytics functions.

5.1.3 FACE RECOGNITION AND QR IMPLEMENTATION

The attendance system implements two verification methods: facial recognition and QR-based verification. Both methods are connected to the same session-based attendance process, while the verification steps differ according to the selected method.

For facial-recognition attendance, the browser camera provides the input and MediaPipe BlazeFace performs the initial face detection on the client side. Suitable frames are then sent to the backend, where InsightFace `buffalo_l` is used for face processing and ArcFace generates the 512-dimensional L2-normalized facial embedding. The captured embedding is compared with the enrolled representations using cosine similarity. Presentation-attack

detection is also performed using MiniFASNetV2 and MiniFASNetV1SE together with the implemented image-based checks. When the required verification conditions are satisfied, the result is passed to the attendance process.

For QR attendance, the implementation uses institutional Google OAuth2/OpenID Connect authentication together with a session-specific QR challenge. The challenge is generated using the session-specific `qr_secret`, HMAC-SHA256, and a rotating 10-second time window. During validation, the backend checks the current or immediately preceding time window, validates the session and section, and checks for duplicate attendance before recording the attendance.

The facial-recognition models are executed through ONNX Runtime. The implementation supports CUDA execution when a compatible GPU is available and provides CPU execution as a fallback. The resulting CPU and GPU performance is evaluated separately in Section 5.2.4.

5.1.4 ANALYTICS IMPLEMENTATION

The analytics functions were implemented using the attendance and academic data maintained by the system. The implementation processes recorded attendance to provide attendance status, trend information, future projection, recovery analysis, and attendance-based what-if analysis.

The system calculates attendance from the recorded session-wise attendance data and uses Ordinary Least Squares (OLS) regression to analyse the attendance trajectory. The resulting trend is used to generate a future attendance projection based on the available attendance history. When a shortage exists, the system also calculates the attendance recovery requirement for reaching the configured target.

The Digital Twin component implements attendance what-if analysis. It allows alternative future attendance conditions to be evaluated and their effect on the projected attendance status to be examined without modifying the actual attendance records.

For academic analytics, the implementation uses the available attendance and assessment data and applies Pearson correlation to examine their observed statistical relationship. The resulting value is used as an analytical measure of association between attendance and assessment results, rather than as evidence of a causal effect.

5.1.5 REPORTING IMPLEMENTATION

The Reporting module was implemented to provide access to attendance and analytical information generated from the records maintained by the system. It uses the existing attendance and academic data rather than maintaining a separate reporting dataset.

The module presents session-wise attendance information for the relevant users. Faculty users can access attendance information associated with their classes and sessions, while students can access their own attendance information and status.

The reporting functions are connected to the same stored records used by the attendance and analytics modules. Therefore, attendance information presented through the reporting functions is derived from the maintained attendance records and can be used alongside the analytical information produced by the system.

5.2 RESULTS

The results of the implemented system are presented through functional attendance testing, recognition and security testing, analytics outputs, and CPU–GPU performance measurements. This section reports the outcomes available from the project's documented testing and benchmark material and does not introduce additional performance or accuracy values.

5.2.1 ATTENDANCE RESULTS

The functional attendance testing evaluates the implemented attendance workflow for the supported QR-based and facial-recognition-based methods. The testing focuses on whether the system accepts valid attendance requests and handles the validation conditions defined in the implementation. The attendance and security interface used during the testing is shown in Fig. 5.1.

For QR attendance, the functional checks cover the session-specific QR validation process, session and section validation, and duplicate attendance handling. For facial attendance, the testing covers the implemented identity-verification process together with the presentation-attack detection checks.

When the required verification conditions are satisfied, the system creates the attendance record for the corresponding student and session. The resulting records are then available to the attendance management and reporting functions. The session and timetable management interface is shown in Fig. 5.2.

Table 5.1 should therefore present the documented functional test cases and their observed results. It should report the actual test outcomes rather than introducing an overall attendance accuracy or success percentage that is not established by the available testing evidence.

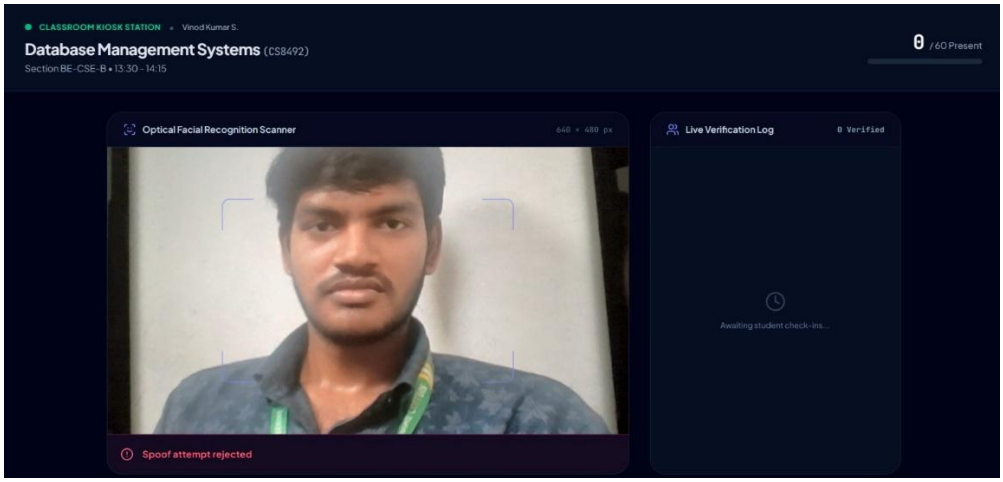


Fig. 5.1. Attendance and Security

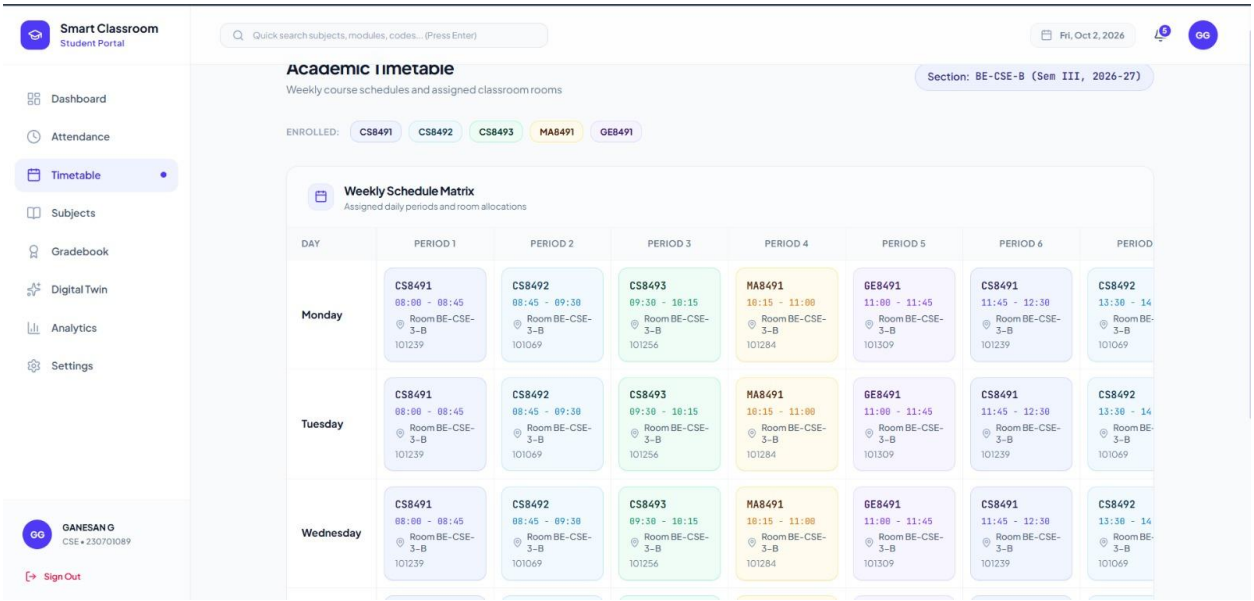


Fig. 5.2. Session and Timetable Management

Table 5.1. Functional Attendance Test Results

TEST CASE	FUNCTION TESTED	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-05	Attendance Session Generation	Create an active attendance session linked to teacher, subject, and classroom	Session created in SCHEDULED status with valid state transition	**PASS**
TC-06	Facial Embedding Storage	Store and retrieve the 512-D facial embedding without loss of vector information	Round-trip cosine similarity = 1.0000000	**PASS**

TC-07	Digital Twin Attendance Projection	Generate future attendance projection from recorded attendance	Projected range = 78.4%–85.2%; risk tier = LOW_RISK	**PASS**
TC-08	Attendance Recovery Analysis	Calculate minimum future attendance required to reach target	19 sessions required from the tested input; feasibility confirmed	**PASS**
TC-09	Attendance Integrity Scoring	Calculate the composite integrity score from verification factors	Score evaluated using the implemented integrity function; tier = ACCEPTABLE	**PASS**

5.2.2 RECOGNITION AND SECURITY RESULTS

The recognition and security results evaluate the implemented verification mechanisms used during attendance marking. The facial-recognition and QR-based methods are discussed separately because their verification processes and test conditions are different.

For facial recognition, the testing covers the implemented face-verification pipeline and presentation-attack detection process. The system performs identity matching using the enrolled facial representations and applies the required presentation-attack checks before the attendance request is accepted. The facial-recognition interface used during testing is shown in Fig. 5.3. The available testing demonstrates the behaviour of these functions under the tested conditions; it should not be interpreted as a general recognition or presentation-attack detection accuracy measurement.

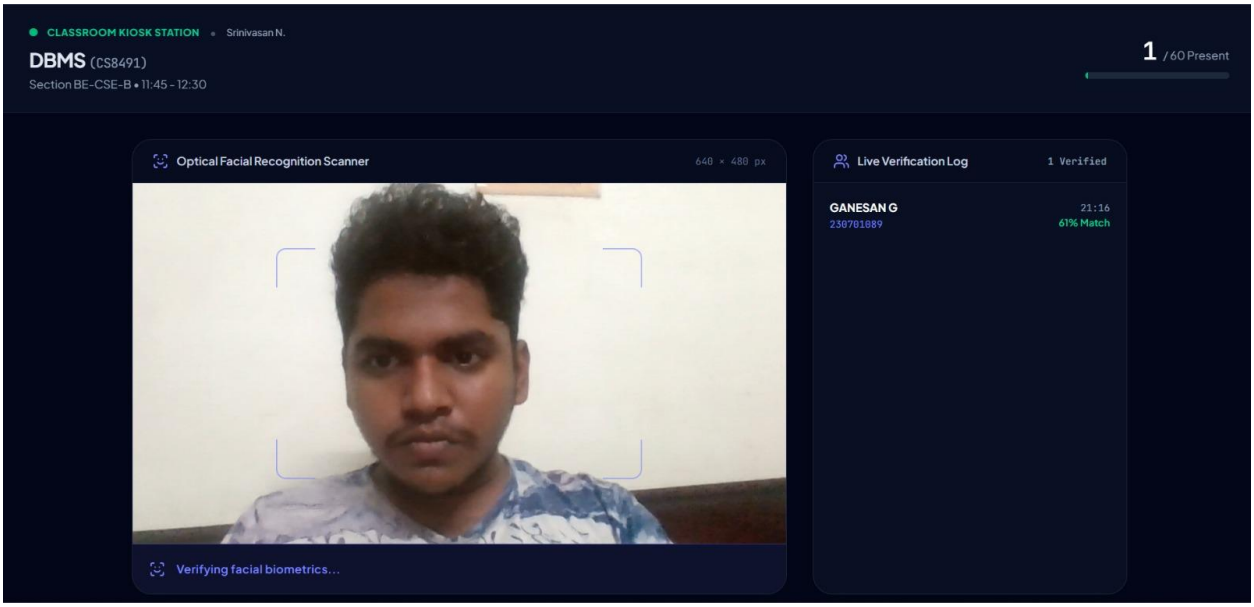


Fig. 5.3. Facial Recognition

For QR attendance, the testing covers the implemented time-bound and session-specific validation process. The QR challenge is checked against the applicable time window and session information before attendance is accepted. The validation process also includes the documented section and duplicate-attendance checks. The QR-based attendance interface used during testing is shown in Fig. 5.4.

The results therefore demonstrate the functional behaviour of the implemented QR recognition and validation mechanisms under the available test conditions. The system accepts attendance only when the required session and validation conditions are satisfied, while invalid, expired, or duplicate requests are rejected. These results do not establish that all possible presentation attacks, QR misuse cases, or other security threats are prevented.

The quantitative CPU and GPU measurements for facial processing are presented separately in Section 5.2.4, where the documented benchmark results are discussed.

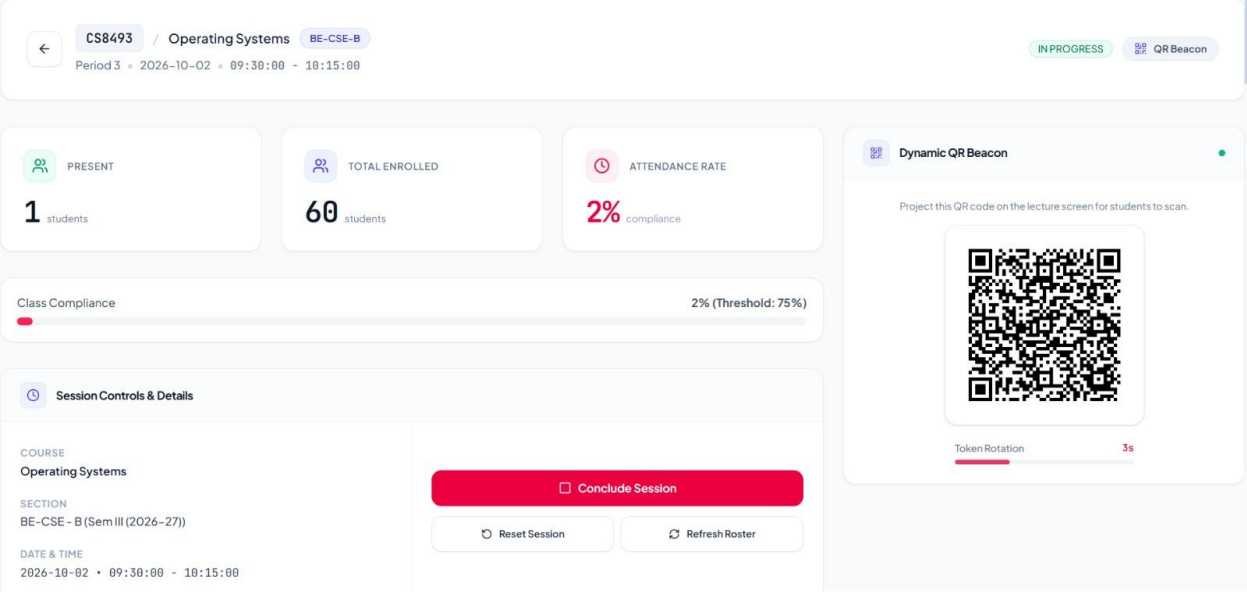


Fig. 5.4. QR Attendance

5.2.3 ANALYTICS RESULTS

The analytics results present the outputs generated from the attendance and assessment data maintained by the implemented system. The analysis covers attendance status, attendance trends, future attendance projection, recovery requirements, Digital Twin-based what-if analysis, and the relationship between attendance and assessment results. The attendance analytics and forecasting interface generated by the system is shown in Fig. 5.5.

The attendance analysis uses the recorded session-wise attendance to determine the current attendance position and identify shortages. The system also applies the implemented trend and projection functions to the available attendance history. Where a shortage is identified, the recovery analysis provides the attendance requirement for reaching the configured target.

The Digital Twin component provides what-if analysis by evaluating alternative future attendance conditions without modifying the actual attendance records. This allows possible changes in attendance status to be examined under different assumed attendance patterns.

The academic analysis combines attendance and assessment data and applies Pearson correlation to examine their observed statistical relationship. The resulting correlation describes the association present in the analysed data and does not establish a causal relationship.

Table 5.2 will summarize the actual analytical outputs documented for the project. Numerical values will be taken directly from the project results rather than inferred or generated.

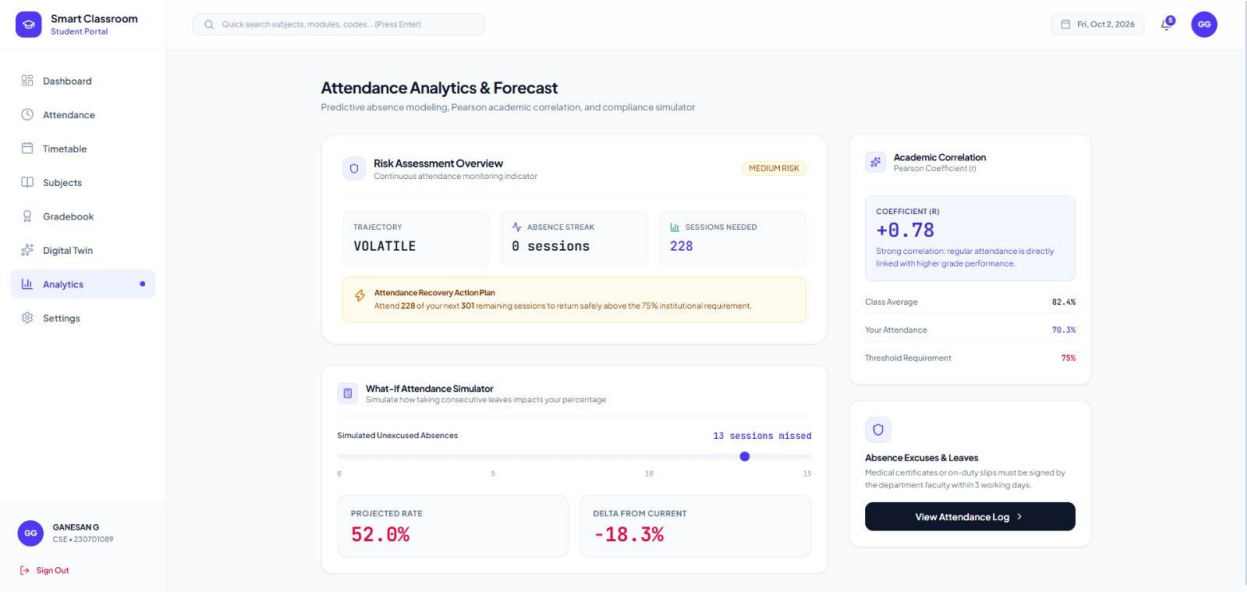


Fig. 5.5. Attendance Analytics

Table 5.2. Summary of Analytics Results

ANALYTICS FUNCTION	TEST INPUT / CASE	RESULT	INTERPRETATION
Attendance Trajectory	Improving attendance sequence	Slope = **+0.1152**; classified as improving	Correctly identified an increasing attendance trend
Attendance Trajectory	Declining attendance sequence	Slope = **−0.1394**; classified as declining	Correctly identified a decreasing attendance trend
Attendance Recovery	13/20 attended, 20 remaining	**19 additional sessions** required; feasible	**95.0%** future attendance rate required
Attendance Recovery	9/20 attended, 10 remaining	**20 sessions** required; infeasible	Maximum achievable attendance = **56.0%**
Attendance – Assessment Relationship	Tested attendance and marks vectors	Pearson **r = 0.9953**	Strong positive association observed in the test data
Digital Twin	25/30 sessions attended	Projected range = **78.4%–85.2%**	Test classified as LOW_RISK

5.2.4 PERFORMANCE RESULTS

This section presents the performance evaluation of the facial-recognition pipeline when executed on a CPU and on a GPU. The comparison is based on the project's documented

benchmark for both detection and recognition stages. The focus is on processing performance under the conditions used in the testing setup.

The system uses ONNX Runtime to run the models. It enables CUDA execution when a compatible GPU is available. If no compatible GPU is found it automatically switches to CPU execution.

Table 5.3 titled "CPU–GPU Facial Recognition Performance Results " shows the benchmark data for both detection and recognition stages as defined in the project's benchmark documentation. All values must be reported exactly as measured using the units and formats specified in the benchmark. No estimated or independently calculated values should be added.

The results compare the two execution modes under the conditions set by the benchmark. The difference observed between CPU and GPU execution reflects the systems behavior with the tested configuration. This difference should not be interpreted as a performance measure for hardware setups or deployment environments.

The benchmark provides evidence of how the facial-recognition pipeline runs. It supports the recognition and security findings discussed in Sections 5.2.1 and 5.2.2.

Table 5.3. CPU–GPU Facial Recognition Performance Results

PERFORMANCE METRICS	CPU	GPU (CUDA)	OBSERVED DIFFERENCE
Mean Face Detection Latency	301.14 ms	16.07 ms	**18.74×**
Mean ArcFace Recognition Latency	1266.49 ms	36.70 ms	**34.51×**

Mean Total Pipeline Latency	1567.63 ms	52.77 ms	**29.70×**
Continuous Video Throughput	0.64 FPS	18.95 FPS	**29.61×**
Identity-Match Agreement	100.0%	100.0%	**No Disagreement**
Mean CPU–GPU Embedding Cosine Parity	—	0.999985	**Near-Identical Vectors**

5.3 DISCUSSION

5.3.1 ATTENDANCE AND SECURITY

The implemented system combines QR-based attendance and facial-recognition attendance into a session-based workflow. Attendance records are only created after the verification process is completed. Each record includes the student, the classroom session, the timestamp and the method used. This setup allows both methods to function together within the attendance management process.

For QR-based attendance the tested cases covered the validation conditions of the implemented mechanism. The current time window and the previous window were accepted. A QR with a window a modified HMAC value or one generated for a different session was rejected. These results confirm that the time-bound HMAC-based and session-specific validation works as intended.

For facial-recognition attendance the pipeline performs face detection, recognition and presentation-attack detection before creating an attendance record. The local evaluation showed a difference between genuine and imposter samples used in recognition testing. The tested anti-spoofing cases also passed. However these results are limited to the samples

and conditions used during testing. They should not be interpreted as recognition accuracy or anti-spoofing accuracy.

The results show that the attendance module provides verification controls while maintaining a consistent session context. Once verification is successful the attendance records are used by the monitoring, reporting and analytics components. This ensures a data source throughout the system.

5.3.2 ATTENDANCE ANALYTICS

The attendance analytics module processes recorded attendance data to deliver trend analysis, future projections, recovery analysis, Digital Twin simulation and academic performance correlation. These functions convert stored attendance records into information for monitoring and further analysis.

Trend analysis uses least-squares regression on the attendance sequence. In the testing an improving sequence gave a slope of 0.1152 while a declining sequence gave a slope of -0.1394. The results were classified according to the implemented trend analysis. This confirms that the calculation responds correctly to changes in the attendance trend.

Recovery analysis calculates the number of sessions needed to reach a target based on the remaining sessions. In a test case with 65% attendance 20 remaining sessions and a 75% target the system calculated that 19 of the remaining sessions must be attended. This corresponds to a required run rate of 95%. A second test case required sessions than were available and the system correctly identified it as infeasible.

The Digital Twin provides what-if analysis by evaluating projected attendance conditions without changing the records. In a test case 25 out of 30 sessions resulted in 83.33% attendance. The projection was reported in the range of 78.4% to 85.2%. Classified as low risk. This demonstrates how the simulation component can evaluate outcomes.

The analysis of academic performance makes use of Pearson correlation with regards to attendance and assessment values. The vectors from the test gave a correlation coefficient value of 0.9953. This means that the computation of correlation is correct when using the test values. However, it does not reflect the real life correlation between attendance and academic performance.

Overall the tested analytics functions worked as intended. They provided trend, projection, recovery, simulation and correlation outputs from the data. The results demonstrate the capabilities of the system without altering the attendance records.

5.3.3 ACADEMIC PERFORMANCE

The academic performance component analyzes attendance and assessment data using Pearson correlation. The goal is to measure the linear association between attendance and academic performance within the dataset.

Correlation computation was checked using the attendance and assessment test vectors that were set out in the problem. The outcome is a value of Pearson correlation coefficient of 0.9953. This value shows the presence of a positive linear relation between the data in the test.

This result is limited to the test vectors used in validation. It does not represent a study of student performance. It also does not prove that attendance causes results. The system presents the correlation as a measure not as evidence that attendance directly determines performance.

This analysis meets the project requirement of linking attendance to performance. It keeps the result separate from any interpretation.

5.3.4 LIMITATIONS

The testing results are based on the test cases and conditions defined for the implemented system. The results show how the system functions within the tested scope. They should not be seen as an evaluation of all classroom conditions.

The facial-recognition evaluation used a set of imposter samples. The anti-spoofing evaluation used a set of fide and attack samples. The reported results verify the behavior of the recognition and presentation-attack detection pipeline for the tested samples. They do not prove accuracy or performance across a population.

The measured facial-recognition performance also depends on the execution environment. The dedicated CPU–GPU benchmark showed differences in detection, recognition and total pipeline latency between CPU and GPU execution. The values in Table 5.3 apply to the benchmark environment. They should not be treated as fixed performance values for every deployment.

The QR security evaluation covered the defined cases: window, previous window expired window, window HMAC tampering and cross-session QRs. These tests confirm the validation behavior for the tested conditions. They do not cover every attack scenario.

The analytics results are also dependent on the input data. The regression, recovery, Digital Twin and Pearson correlation tests used defined test cases and test vectors. The Pearson result confirms the calculation for the supplied data. It should not be interpreted as evidence of a causal link, between attendance and academic performance.

In summary the current evaluation shows that the implemented modules work as expected within the tested scope. To evaluate performance and generalization broadly larger and more varied datasets additional classroom conditions and more deployment environments would be needed.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The project presents the implemented Smart Classroom and Autonomous Attendance Management System offering automatic classroom attendance verification with attendance information for its monitoring and analysis. The project tackles the identified issues by introducing the automatic verification of attendance with management, reporting, and analytical features for monitoring.

The implemented attendance procedure provides QR-based and facial recognition procedures within a unified session management for classroom attendance. QR attendance was introduced with institutional authentication and a time-sensitive, session-specific challenge. The facial attendance procedure was implemented with facial recognition and presentation attack detection, with attendance information tied to the relevant student and classroom session. The information was then stored in a PostgreSQL database for later monitoring and analysis.

The testing procedures confirmed the proper functioning of the implemented attendance procedures and security features within the identified constraints. QR scanning correctly distinguished the current and previous valid attendance windows while rejecting the tested invalid, expired, future, modified, and cross-session attendance windows. The facial recognition procedure and presentation attack detection were validated with the provided test examples. The results illustrate the performance of the implemented pipeline for the tested cases and do not imply general functionality or security.

The attendance information was then used to implement the analytical features including trend analysis, future projection, recovery analysis, Digital Twin what-if analysis, and Pearson’s correlation with the assessment information. The validation procedures

confirmed the correct performance of the calculations for the specified analytical procedures with the provided test cases and test vectors. The Pearson's correlation result illustrates the observed value dependency for the provided vector and should not be used for implying causality between attendance performance and assessment results.

The project then proceeds to evaluate the performance of the facial recognition pipeline under CPU and GPU execution. The performance benchmark was used to illustrate detection, recognition, and overall pipeline latency with the tested hardware, with lower values observed for the GPU under the same test conditions. The results illustrate the performance differences for the tested conditions but should not be used as a general reference for all CPU and GPU combinations.

In summary, the project implements the automated attendance verification and management system with monitoring and analytics for classroom use. The presented features and test results confirm the implemented procedures within the project description for the identified constraints. The limitations identified in the testing procedures can be used for further development and testing within the presented application.

6.2 FUTURE WORK

The potential future direction of work on the system can consist of expanding the evaluation and deployment of the implemented attendance, verification, security, and analytics components. The following areas are directly connected to the described limitations and can be considered for future work.

1. Expanded facial-recognition evaluation

The facial-recognition and presentation-attack detection components can be evaluated using more extensive and diverse datasets. The utilization of such data would allow for a more significant and varied validation of the components' functions than the one allowed by the current state of datasets.

2. Broader hardware and deployment evaluation

The facial-recognition pipeline can be evaluated on more CPU and GPU classes and deployment environments. Such an expansion would allow for a more significant validation of the pipeline's behavior against the current CPU–GPU benchmark.

3. Extended QR security testing

The QR attendance mechanism can be evaluated against more extensive security and edge-case testing beyond the already-implemented time window, HMAC modification, future challenge, and cross-session challenge validation. This would allow for a more significant evaluation of the QR verification mechanism against various edge cases and attacks.

4. Larger-scale analytics validation

The attendance trend, recovery, Digital Twin, and academic-performance analysis can be validated against more extensive real-attendance and assessment data. This would allow for a more significant validation of the analytics functions beyond the already-implemented test cases and test vectors.

5. Wider system evaluation

The application as a whole can be evaluated on larger class sizes and institutional scales to observe the functions' behavior against a larger volume of data and concurrent attendance and analytical operations. This would expand the current utilization of functional testing and system performance benchmarking to a wider scale than a single class or educational institution.

In general, the aforementioned areas of future development represent the potential directions of the project beyond the current state. However, they do not involve any significant changes to the application's architecture and utilize implemented functions, which can be considered as the basis for further, more complicated additions and alterations

PUBLICATIONS

1. Ganesan G, Gokula Sarathy P S, Harshavardhanan V, Himeshwar N and Sabitha R, “A Multi-Modal Student Attendance System with Face Recognition and Time-Bound QR Authentication”, Submitted in *8th IEEE International Conference on Emerging Electronics (ICEE 2026), Banglore, India.*

2. Ganesan G, Gokula Sarathy P S, Harshavardhanan V, Himeshwar N and Sabitha R, “A Multi-Modal Student Attendance System with Face Recognition and Time-Bound QR Authentication”, Submitted in *7th International Conference on Computing, Communication, and Industry Standard 6.0 (C2I6-2026), CMRIT Banglore, India.*

A Multi-Modal Student Attendance System with Face Recognition and Time-Bound QR Authentication

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***Abstract*— Attendance is an important component of academic management. However, manual attendance may cause loss of valuable class time, errors, and proxy attendance. Face recognition and QR-based methods can mitigate these issues; however, face recognition can be affected by presentation attacks, and QR codes may be shared between students. This work proposes a multi-modal smart classroom attendance and academic management system that uses both face recognition and time-bounded QR authentication. The face module performs face detection, quality filtering, facial embeddings, and presentation-attack detection, whereas the QR module utilizes a rotating HMAC-SHA256-protected challenge, institutional Google authentication, and session/section validation. The proposed platform consists of teacher and student portals, timetable and session management, manual attendance, attendance integrity scoring, analytics, digital-twin based analysis, and assessment/marks management. The developed system was implemented as a web-based platform. It was evaluated using valid and invalid attendance scenarios, and face-processing performance benchmarking. The obtained results highlight the feasibility of employing multi-modal approaches for attendance management supplemented by academic management features in a single platform.**

Keywords— Smart classroom, student attendance, face recognition, QR authentication, anti-spoofing, HMAC-SHA256, attendance analytics, academic management

I. Introduction

Attendance is a critical element of academic management, aiding on both the monitoring of student attendance and the generation of reliable academic records. Many classrooms track attendance manually by roll calls or written records. While this maintains control by the instructor, it is often time-consuming during teaching activities and can lead to errors, omissions and proxy attendances. This has led to the adoption of alternative automatic approaches including QR codes, RFID and face recognition. Previous works have demonstrated the benefits of such approaches, but each have limitations.

Face recognition is advantageous to students in that they do not have to actively engage with the system in taking attendance. However, this technology can be limited by lighting conditions, face position, image quality and the distance to the camera. Another concern is that of presentation attacks; a photograph or other representation of a student can be used to impersonate their characteristics. QR-based attendance is convenient in that it only necessitates a smartphone device and can be used as a substitute when face recognition is not viable. However, QR codes shown in a classroom can be photographed and shared, requiring additional steps for verification.

This work presents a multi-modal smart classroom attendance and academic management system which combines face recognition and time-bound QR- authentication. The face-based module consists of face-detection, quality filtering, facial embeddings, and presentation-attack detection prior to recording attendance. The QR module consists of a rotating time-bound challenge protected by HMAC-SHA256 plus institutional Google authentication and session/section validation for a variety of authentication options appropriate for classroom settings.

The system is also designed to support for other related activities around attendance management. It offers separate teacher and student portals, classroom and session management, timetable and subject information, manual attendance, attendance reports, integrity scoring, attendance analytics, digital-twin-based attendance analysis, and assessment and marks management. This way, attendance data can be maintained and analyzed as an enabler of wider academic management rather than a standalone record.

It is the aim of this work to unify attendance, authentication, classroom management, and academic management in a single platform. Novelty claims are not made for the technologies presented individually. Instead, this work focuses on their practical application in the classroom and the security considerations involved.

II. LITERATURE SURVEY

A. Face-Recognition-Based Attendance

Face recognition has been widely studied for classroom attendance. Bhattacharya et al. introduced the Smart Attendance Monitoring System (SAMS) that utilizes face detection, face quality, and recognition to automate attendance in a classroom setting [1]. They note the importance of selecting appropriate facial images before performing recognition. Chintalapati and Raghunadh designed an automated attendance system that consists of image acquisition, face detection, preprocessing, feature extraction, classification, and post-processing steps. Their study discusses spoofing as a security concern for face-based attendance systems [2]. Dev and Patnaik studied a face-recognition attendance system that used various recognition approaches and evaluated performance under varying illumination, head movement, facial expressions, and camera distance [3]. These studies show the potential for contactless attendance while also illustrating how recognition performance can be affected by imaging conditions.

Further studies have examined the use of real-time operation in classes. Sultan et al. introduced a real-time university classroom attendance system, which used an HD camera, HOG-based facial feature detection, and the Dlib face-recognition API. Their system was able to recognize multiple students from different viewpoints and achieved an accuracy of 97.38% under their experimental conditions [4]. Lateef and Kamil designed a smart-classroom attendance system with YOLOv7 and a database of 31 students, comparing approaches of machine-learning and deep-learning-based face recognition [5]. Raj et al. also introduced a face-recognition-based smart attendance system to reduce the need for manual attendance [6]. These studies illustrate the shift from traditional face-recognition pipelines to those that allow for real-time and automated attendance in the classroom.

The use of multimodal verification has also been investigated. Akbar et al. combined face recognition and RFID verification for classroom entry and attendance, using facial identities along with an RFID-based identification card [7]. Their system demonstrates the use of multiple independent identification mechanisms within a single attendance procedure. However, such systems entail additional hardware requirements and implementation complexity.

B. QR-Code-Based Attendance

QR codes have been considered as an alternative solution since students can use mobile devices to check their attendance. Liew and Tan proposed a QR-code-based attendance management system that verified the class time, registered mobile devices, and geolocation in addition to QR code [8]. Their findings showed that contextual information could be used to enhance such solutions in order to prevent attendance submissions from unauthorized locations of users. Wiriasto et al. introduced a QR code-based Android attendance application using Flutter that is operated by different features intended for students and lecturers, including geolocation QR-code scanning [9]. Bacuna and Dadiz introduced an eSAM server and mobile QR attendance system that recorded the course name, section, subject, date, and attendance information while providing summary reports to instructors and administrators [10].

In turn, other researchers investigated QR-based attendance management systems that included other tracking and management procedures. Casunuran et al. presented a QR attendance system that included GPS-based student tracking and SMS notification for the arrival and departure of students [11]. Masalha and Hirzallah introduced a QR-based attendance management system for student attendance recording using QR identification [12]. Nuhi et al. developed a QR-based web attendance management system aimed at helping attendance recording and providing automatically generated information on attendance [13]. Pati et al. introduced a QR-based tracking system where personalized QR codes are used by students while their entry and exit information are recorded [14]. Ambaye et al. more recently developed and evaluated an automated QR attendance system for classroom management, reporting improved record keeping, decreased paperwork, and user acceptance [15]..

C. Research Gap

The reviewed literature shows that face recognition can be used to automate student identification, while QR-based systems can ease mobile attendance recording. Some works also included other mechanisms, e.g., RFID checks [7], registered devices and geolocation [8], or location tracking and notifications [11]. Yet, the systems reviewed do not show the same combination of face recognition, presentation-attack detection, time-bounded cryptographically protected QRs, institutional identity checks, and unified classroom and academic management that this work introduces.

As such, the contribution of the proposed system is not a new face-recognition algorithm, QR technology, or cryptography: the originality lies in unifying different attendance mechanisms and the associated security checks in a single classroom platform where face recognition and QR authentication can be used as alternatives and share the same attendance-management infrastructure.
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III. PROPOSED SYSTEM

The proposed system is a smart classroom attendance system aimed at minimizing the efforts related to manual attendance while also maintaining the integrity of the attendance process. Several attendance modes are allowed including QR attendance, face recognition attendance, QR and face recognition attendance, and manual attendance, all housed in one software. Attendance is tied to classroom sessions and timetable, thus enabling reports generation, integrity checks and academic analytics.

A. System Architecture

The system is designed using a client-server model with React.js as the frontend framework plus Django REST

framework as the backend framework. The system has two interfaces, the teachers' and students' interfaces. The backend comprises authentication, classroom sessions, attendance, face data, timetable, and student records.

Initially, the teacher logs in to the system and creates a session for the class he/she wants to take. The teacher selects his/her preferred method of taking attendance, thus initiating the verification process. This involves verifying the students' faces depending on the mode selected by the teacher. The system's backend then validates and authenticates the information provided and registers the attendance of the students in the created session. The records are thereafter subjected to integrity tests, reports generated, and analytics done as explained in Fig.1 .

Authentication → Session creation → Mode selection → Student verification → Validation and authentication → Attendance recording → Integrity analysis → Analytics.

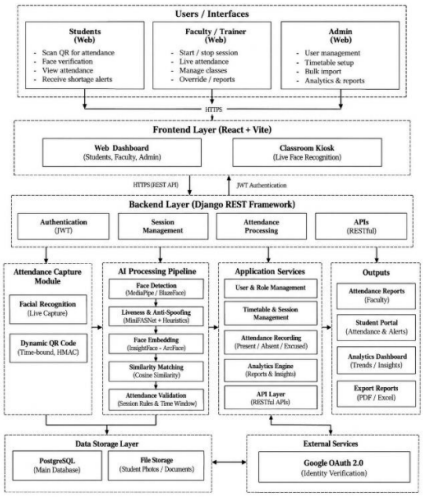


Fig. 1: System Architecture

Module	Main Function
QR Attendance	Time-bound QR challenge and institutional authentication
Face Attendance	Face detection, embedding generation, and identity matching
Presentation-Attack Detection	MiniFASNet ensemble and image-based rejection checks
Attendance Integrity	Verification metadata and 0–100 integrity scoring
Session Management	Class session creation, activation, and attendance control
Academic Management	Subjects, assessments, marks, and student records
Analytics	Attendance summaries, shortages, trends, and Digital Twin analysis

Table I — System Modules and Functions

B. QR-Based Attendance

QR module provides a simple QR code-based attendance recording mechanism with a smartphone without displaying QR code permanently. When a QR session is initiated, the backend generates a unique session token and secret key to create a QR challenge with a HMAC-SHA256 encrypted value with the timestamp window.

Any captured QR codes are invalidated by a new challenge generated every 10 seconds. Students authenticate by logging in with their institutional Google account and scanning the QR code. The backend validates the identity token, matches the student with the corresponding QR session, and records the attendance after successful challenges and Google authentication, as depicted in Fig. 2.

Besides the QR scanning step, the proposed solution offers multiple levels of protection to prevent unauthorized marking of attendance. Simultaneously, it ensures that a student is not marked present more than once for the same session.

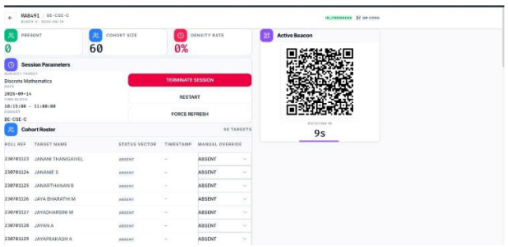


Fig. 2: Dynamic Qr Attendence

C. Face Recognition and Presentation-Attack Detection

This face module is meant to replace QR check-in without touching anything. A camera in the room looks at the student. Then MediaPipe runs the first face search. It keeps only frames that meet simple rules like detection confidence, face size, and where the face sits in the image.

After that, the kept frames go through the InsightFace buffalo_l** setup. From each usable frame, the system makes a feature vector with 512 values. Those vectors are scaled with an L2 step. Next, the system compares each vector to the stored vectors for students in the same section. It uses cosine similarity for that match. The system logs attendance only if the match passes the identity test.

Face matching alone cannot prove that a real person is in front of the camera. A spoofing check is added for that reason. The method uses MiniFASNetV2 and MiniFASNetV1SE together as an ensemble. They estimate if the face looks live. After that, the system runs four quick image tests to reject likely attacks. These are Laplacian texture checking, Canny edge density checking, FFT based moiré checks, and Hough based device border checks. Attendance is accepted only after the system sees a run of valid frames. The full face work flow is shown in Fig. 3.



Fig. 3: Anti-Spoofing in Face Recognition.

D. Attendance Integrity

The system stores for each check an attendance state plus additional information. Each student's record contains within a session their ID, time of check, method of verification, face confidence, and additional verification notes.

The attendance state can be PRESENT, ABSENT, LATE, or EXCUSED.

A percentage between 0 and 100 is calculated for each check based on such factors as QR validity, face recognition confidence, anti-spoofing confidence, a check's temporal validity, and the verification path taken. Notes about the score and general verification notes are attached to the check, allowing the teacher to assess the reliability of each attendance record and draw attention to potentially suspicious records.

E. Classroom and Academic Management

The attendance system is integrated with timetable and session management to enable recording of attendance against the relevant class. Through the teacher portal sessions can be managed, attendance rosters and reports viewed and reviewed, integrity information viewed, assessments managed and authorized attendance corrections made.

The student portal provides access to attendance history, timetable, subjects, grades, and personal analytics. This ensures that attendance information is accessible to the students outside the classroom providing a better overview of students' academic standing.

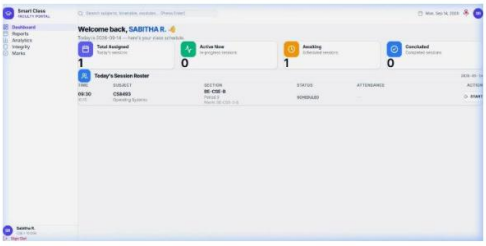


Fig. 4: Faculty/Teacher Page

F. Attendance Analytics and Digital Twin

The attendance data can be further processed to produce student- and class-based attendance summaries, find attendance shortages, and analyze attendance trends within sessions and subjects. Additionally, the system utilizes a Digital Twin to analyze trends in attendance history and determine whether the trends are improving, deteriorating, or stable for a given student.

The module is then able to estimate how many more sessions a student would need to attend to meet the required level of attendance and, subsequently, perform a what-if analysis depending on the projected future attendance levels. Similarly, attendance data can be combined with assessment results to evaluate risk levels.



Fig. 5: Student Portal and Analytics

Conclusively, as seen in the proposal, the system provides attendance verification, classroom session control, attendance audit processes, academic record storage, and analytical tools in a single system. It must be stressed that the main innovation of this project is not in the creation of new algorithms or cryptographic protocols but in the unification of the aforementioned technologies into a singular smart classroom attendance system.

IV.METHODOLOGY & IMPLEMENTATION

The system was implemented as a Web application using the React frontend and Django REST Framework backend. QR authentication, face recognition, attendance verification and the academic services were unified around the session-based attendance management model.

A. Technology stack

The frontend was built using React 18 and Vite while the backend was using Django 5.x and Django REST Framework. The REST APIs unified the system's frontend and backend parts. SimpleJWT was used to provide the application's authentication while the institutional Google authentication was used during the QR-based attendance sessions. SQLite was used for the development while the production used the database. The face processing part used MediaPipe to detect the faces on the client-side, while InsightFace buffalo_l for the face embeddings, while NumPy and SciPy were used for the analytics module.

B. QR authentication

When the teacher begins the QR session, the backend generates a unique session token and a random session secret which are used to build a temporary challenge with the session identifier, time-window and HMAC-SHA256 digest.

QR-based attendance uses a dynamic 10 seconds' time-window which renews itself periodically. When a student scans a QR code, the server validates the challenge, the time-window and decodes the Google institutional ID token to check against the known user profile and attendance session. This way, the session, student and QR code are bound together during the scan to provide additional security. Several verification steps are executed instead of relying on the single QR code scan.

C. Face recognition

MediaPipe BlazeFace is used to initialize the face detection from the camera feed, which is subsequently filtered using the confidence threshold, face size and position on the screen before being subject to recognition. This way, the irrelevant video frames can be discarded earlier the pipeline to reduce the processing overhead.

The detected face is processed using the InsightFace buffalo_l model to extract a 512D L2-normalized embedding, which is compared to the known student embeddings using the cosine similarity. A configured cosine threshold is used to determine the match during the recognition. Additionally, MiniFASNetV2 and MiniFASNetV1SE are used to estimate the liveness of the face with the live probability averaged between the two models. Four image-based checks are also used during the liveness detection to discard non-living faces at an early stage: the texture analysis using the Laplacian filter, Canny edge density analysis, FFT-based moiré analysis and Hough-based device border detection. Finally, three consecutive verification frames need to be successfully processed before the result is accepted.

D. Attendance integrity and system integration

The attendance record binds the student, session, status, timestamp, method, face confidence, and verification metadata together into a single record with an additional integrity score between 0 and 100. The score is computed from the QR validation, face recognition confidence, anti-spoofing confidence, temporal consistency of the timestamp, and the method of verification. The backend Application Programming Interfaces (APIs) are organized around the authentication, sessions, attendance, QR verification, recognition, academic records and analytics. The attendance management follows the teacher and student roles, with the teacher managing the QR sessions, and the student having the view-only access. Additionally, attendance binds the timetable information and the session data together, which allows the system to unify the teacher reports, student history, attendance analytics and Digital Twin analysis.

Overall, the implementation focused on unifying the verification methods around the attendance management and providing additional security during the both QR and face recognition steps, while maintaining the connection to the academic records.

V. RESULTS AND LIMITATIONS

The prototype was evaluated in order to verify the correctness of the described attendance procedures, rejection of invalid requests, and behavior of the face-recognition pipeline. However, limited resources were devoted to biometric evaluation and thus most results are based on functional and security testing rather than statistical analysis of recognition or presentation-attack detection rates.

A. Functional Evaluation

The QR attendance procedure was tested on valid and invalid authentication attempts. A valid request was a valid institutional account with a valid QR challenge. Non-institutional accounts, accounts from another section, invalid QR challenges, inactive sessions, and repeated attendance were rejected. These tests demonstrate that the attendance procedure correctly implements identity, session, section, and attendance-duplication checks on the backend.

B. QR Authentication Evaluation

The QR attendance procedure was tested on altered QR codes in order to evaluate the security of the QR parsing and HMAC verification procedure. Altered QR fields such as the session, time window, and HMAC signature resulted on failed attempts. QR challenge generation is based on a session HMAC secret and time window on the server side while institutional Google accounts serve as an additional authentication factor.

QR code rotation is designed to occur every 10 seconds which limits the usefulness of a compromised QR code. These tests demonstrate that the QR attendance procedure resists casual tampering attempts and the presence of an institution account or a valid QR code is not sufficient for successful authentication.

C. Face Recognition and Presentation-Attack Evaluation

The face recognition procedure was tested as a combined procedure of simultaneous face detection, frame filtering, face embedding generation, face matching, and presentation-

attack detection. The prototype uses MediaPipe for face detection and InsightFace for face embedding generation and comparison to the enrolled student embeddings. Four image-based presentation-attack detection checks are implemented using MiniFASNetV2 and MiniFASNetV1SE with additional temporal checks to filter out frames with consecutive valid face detection.

A single valid face frame is not sufficient for attendance due to additional temporal checks. However, the prototype does not include a benchmark presentation-attack detection dataset for evaluating the false acceptance and false rejection rates. Similarly, the recognition accuracy, false acceptance, and false rejection rates cannot be reported for the prototype due to the lack of a controlled benchmark dataset. Thus, the current evaluation should be considered a functional test of the described face-recognition pipeline.

D. Face-Processing Performance

The face-processing performance was previously reported in the IEEE submission as 180-250 ms/frame for GPU execution and 5000-6000 ms/frame for CPU execution. However, the technical audit of the current prototype code did not identify any benchmark dataset or measurements that could be used to reproduce these results. The code includes time measurements of individual components but no timing statistics or large-scale performance benchmarks.

Thus, the performance of the face-recognition procedure in the IEEE submission cannot be reproduced and these performance indicators should be omitted from the public version. Instead, it is possible to report that face recognition using the GPU is significantly faster than the CPU for all components which can be seen in the comparison of the measured values.

E. Discussion

The evaluation demonstrated that the described system architecture allows for the implementation of multiple attendance procedures in a unified classroom platform. QR attendance procedure provides active institutional account authentication with additional session and time-window checks while face attendance procedure serves as a convenient contactless alternative with presentation-attack detection.

Instead of providing strong statistical evidence of recognition or presentation-attack detection performance, the evaluation presented the functional validity of the described approaches to multi-modal attendance. This conclusion is based on the fact that the prototype has not been benchmarked against a large-scale biometric dataset and cannot provide reliable statistical evidence of recognition performance at this time.

The main contribution of this project is the combination of these two attendance procedures into a single classroom platform rather than the performance characteristics of either one.

F. Limitations

The current prototype does not include a large-scale biometric dataset for evaluating the recognition accuracy, false acceptance rates, false rejection rates, and presentation-attack detection rates. Additionally, the current face-enrollment procedure does not include enough reference samples for each student for accurate recognition. Thus, the validity of the prototype depends on the student's appearance during the face detection procedure.

Changes in lighting, occlusion of the face, change of facial expression, and distance between the camera and the student can reduce the accuracy of the prototype. Similarly, the cosine-similarity threshold that determines whether a face matches a reference has not been calibrated using a large-scale benchmark dataset. The presentation-attack detection procedure similarly has not been benchmarked against a standard benchmark dataset.

The QR attendance procedure has not been stress-tested across multiple classrooms while the current SQLite-based prototype has not been optimized for production traffic. The code-base contains legacy and new attendance procedures while the complete end-to-end QR_AND_FACE procedure has not been tested.

Test Case	**Expected Result**	**Observed Result**
Valid institutional account with valid QR	Attendance accepted	Passed
Non-institutional Google account	Attendance rejected	Passed
Student from incorrect section	Attendance rejected	Passed
Valid QR from active session	Attendance accepted	Passed
Modified or invalid QR challenge	Attendance rejected	Passed
Invalid or closed session	Attendance rejected	Passed
Repeated attendance submission	Duplicate prevented	Passed

Table II. Functional Evaluation Of Qr Attendance

VI. FUTURE WORK AND CONCLUSION

A. Future Work

The future work includes enhancing the system's reliability, evaluation, and scalability. A large and diverse face dataset can be used to evaluate the recognition and presentation-attack mechanisms with different lighting, poses, occlusions, and classroom conditions. Additionally, the threshold of the face matching can be calibrated between genuine and impostor samples.

The QR_AND_FACE workflow needs to undergo more extensive evaluations in multiple classrooms and users to determine its practical performance and scalability. The same applies to the QR_AND_FACE workflow in terms of combined verification mechanism. Apart from that, other possible improvements include enhanced attendance audit logging, optimized face processing, and database migration from the current development database to production-grade database.

B. Conclusion

This paper proposed a smart classroom attendance system that integrates QR-based authentication, face recognition with presentation-attack detection, attendance integrity assessment, and academic analytics into one platform. Unlike the existing solutions that focus only on the mechanisms to record attendance, this study demonstrates how attendance records are connected to classroom sessions, timetable, academic records, and student analytics.

The functional tests revealed that QR workflow is able to accept a valid attendance request and reject a few invalid or unauthorized requests. Similarly, the face-based component can be used as a contactless alternative that is empowered by the presentation-attack detection mechanism. The proposed system is also expanded through teacher and student portals, attendance reports, shortage analytics, and Digital Twin-based insights.

The current proof of concept demonstrated that it is possible to combine all these components in one classroom platform. However, the absence of a large labelled biometric dataset and extensive evaluations in a controlled environment prevent the performance analysis on a larger scale. The future work will include additional validation in different classrooms and users to ensure the system's practical reliability.

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Abstract:
Attendance is an important component of academic management. However, manual attendance may cause loss of valuable class time, errors, and proxy attendance. Face recognition and QR-based methods can mitigate these issues; however, face recognition can be affected by presentation attacks, and QR codes may be shared between students. This work proposes a multi-modal smart classroom attendance and academic management system that uses both face recognition and time-bounded QR authentication. The face module performs face detection, quality filtering, facial embeddings, and presentation-attack detection, whereas the QR module utilizes a rotating HMAC-SHA256-protected challenge, institutional Google authentication, and session/section validation. The proposed platform consists of teacher and student portals, timetable and session management, manual attendance, attendance integrity scoring, analytics, digital-twin based analysis, and assessment/marks management. The developed system was implemented as a web-based platform. It was evaluated using valid and invalid attendance scenarios, and face-processing performance benchmarking. The obtained results highlight the feasibility of employing multi-modal approaches for attendance management supplemented by academic management features in a single platform.

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APPENDICES

Appendix A — Selected Functional Test Results

The following selected test cases provide supporting evidence for the functional behaviour discussed in Chapter 5.

TEST ID	MODULE TESTED	TEST CONDITION	RESULT
TC-01	User Authentication	Valid user authentication and token generation	**PASS**
TC-02	Classroom Authentication	Valid classroom authentication and rejection of an incorrect password	**PASS**
TC-03	Timetable Generation	Timetable generated with defined class duration and lunch-break constraints	**PASS**
TC-04	Bulk Import	Valid record accepted and malformed record rejected through validation	**PASS**
TC-05	Session Generation	Attendance session generated for a defined timetable slot	**PASS**
TC-06	Recognition Storage	512-dimensional facial embedding stored and retrieved with cosine similarity of **1.0000000**	**PASS**
TC-07	Digital Twin	Projection performed for **25 attended sessions out of 30**	**PASS**

TC-08	Recovery Analysis	Required attendance calculated for a **75% target** with 20 sessions remaining	**PASS**
TC-10	Academic Data Integrity	Duplicate registration rejected by the database uniqueness constraint	**PASS**

Appendix B — CPU–GPU Benchmark Details

The following results provide supporting detail for the CPU–GPU performance comparison presented in Section 5.2.4. The dedicated benchmark evaluated five real photographic samples using the tested facial-recognition pipeline under CPU and GPU execution.

SAMPLE	CPU TOTAL LATENCY (MS)	GPU TOTAL LATENCY (MS)	SPEEDUP
Aug 21 Image	1643.05	49.34	**33.30**×
Aug 24 Image	1450.20	50.31	**28.83**×
Aug 27 Image	1739.17	50.70	**34.31**×
Final TCS Photo	1543.45	56.18	**27.47**×
Profile	1462.29	57.34	**25.50**×

The benchmark reported the following mean performance values:

PERFORMANCE MEASURE	CPU	GPU (CUDA)
Detection Latency	301.14 ms	**16.07 ms**
Recognition Latency	1266.49 ms	**36.70 ms**
Total Pipeline Latency	1567.63 ms	**52.77 ms**
Continuous Throughput	0.64 FPS	**18.95 FPS**